$$v^2 \times l = c^2 \times m \times h_f$$

$$\therefore h_f = \frac{v^2 l}{c^2 m}$$

Loss of head due to friction, $h_f = \frac{v^2 l}{c^2 m}$

Example – 4

A 300 mm diameter and 600 m long pipe connects two reservoirs. The difference in water levels between the reservoirs is 3 m. Determine the velocity of flow in the pipe. Take Chezy's constant as 60.

Given data:

Diameter of pipe (d) = 300 mm = 0.30 m
Length of pipe (l) = 600 m
Difference in water level (h_f) = 3 m
Chezy's constant (c) = 60

To determine:

Velocity of flow in the pipe (v)

Solution:

Velocity of flow, $v = c\sqrt{mi}$

Hydraulic mean depth,

$$m = \frac{d}{4} = \frac{0.30}{4} = 0.075m$$

Hydraulic gradient,

$$i = \frac{h_f}{l} = \frac{3}{600} = 0.005$$

$$\therefore Velocity, v = c\sqrt{mi}$$

$$= 60 \times \sqrt{0.075 \times 0.005} = 1.162 \, m/s$$

Result:

Velocity of flow, v = 1.162 m/s.

Example – 5

Water flows through a pipe, 50 mm diameter and 20 m long with a velocity of 3 m/sec. Find out the loss of head due to friction. Take Chezy's constant as 60.

Given data:

Diameter of pipe (d) = 50 mm = 0.05 m
Length of pipe (l) = 20 m
Velocity of flow (v) = 3 m/s
Chezy's constant (c) = 60

To find out:

Loss of head due to friction, h_f

Solution:

We know, Chezy's formula for loss of head in terms of velocity $h_f = \frac{v^2 l}{c^2 m}$

Hydraulic mean depth, $m = \frac{d}{4} = \frac{0.05}{4}$

$= 0.0125$ m

Head loss due to friction,

$$h_f = \frac{v^2 l}{c^2 m}$$

$$= \frac{3^2 \times 20}{60^2 \times 0.0125} = 4 \text{ m}$$

Result:

Loss of head due to friction, h_f = 4 m.

7.6 Pumps

7.6.1 Definition

A pump is a device that moves fluids (liquids or gases) or sometimes slurries by mechanical action. There are two basic classification of pumps. They are,

- Positive displacement pumps
- Rotodynamic pumps

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